

AI AND IOT-ENABLED SMART OFFICE SYSTEM WITH CLOUD-BASED ENERGY OPTIMIZATION

25-26J-034

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in Computer

System & Network Engineering

Department of Computer System Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**THE AI-BASED SMART LIVE EMPLOYEE STRESS
MANAGEMENT SYSTEM, AI-BASED EMPLOYEE CURRENT
REPORT GENERATOR SYSTEM USING CAMERA
AND SMART TASK SCHEDULING & MONITORING SYSTEM**

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Declaration

Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the Supervisor:

(Prof.Anuradha Jayakody)

Date: 8/29/2025

Abstract

An AI and IoT-Enabled Smart Office System with Cloud-Based Energy Optimization is proposed in this project with the goal of improving operational efficiency, worker well-being, and workplace productivity. An AI-Based Employee Current Report Generator, which uses IoT-enabled cameras and AI algorithms to automate activity tracking and reporting; an AI-Based Smart Live Employee Stress Management System, which uses sensors and machine learning models to continuously monitor employee stress levels for proactive interventions; and a Smart Task Scheduling & Monitoring System, which uses IoT data and AI-driven analytics to intelligently assign tasks, track progress, and optimize workload distribution, are the three intelligent modules within the system. This methodology is new in contrast to traditional ways since it allows for cloud-based resource optimization while offering real-time, data-driven insights for job management and staff wellbeing. Advanced AI approaches, IoT integration, and cloud analytics for decision-making are all part of the proposed system's goal to create a smart workplace environment that is balanced, effective, and sustainable.

Keywords: AI-enabled office system, IoT integration, real-time stress management, automated reporting, smart task scheduling, workplace productivity, employee well-being, machine learning, smart office technology.

Table of Contents

Contents

Declaration	3
Abstract	2
Table of Contents	3
List of Figures	4
List of Tables.....	4
List of Abbreviations	4
1.0 INTRODUCTION.....	5
1.1 Background and Literature	5
1.2 Research Gap.....	7
1.3 Research Problem.....	8
2.0 Objectives.....	9
2.1 Main Objective	9
2.2 Specific Objectives	9
3.0 METHODOLOGY	10
3.1 Requirement Gathering	10
3.1.1 Past Research Analysis	10
3.2 Feasibility Study.....	12
3.2.1 Technical Feasibility.....	12
3.2.1.1 Knowledge on Technologies	12
3.2.1.2 Knowledge on Tools	14
3.2.1.3 Data collection Knowledge	17
3.2.2 Schedule Feasibility	19
3.3 System Analysis	20
3.3.1 Software Solution Approach.....	20
3.4 Project Requirements.....	23
3.4.2 Non-Functional Requirements.....	23
3.5 Testing.....	24
3.6 Timeline	25
4.0 Budget	27
5.0 SUMMARY	28
6.0 References	29

List of Figures

Figure 1 - task scheduling & monitoring system flowchart.....	21
Figure 2 - Live employee stress management system	21
Figure 3- current report generator system flowchart	21
Figure 4 - work breakdown structure.....	22
Figure 5 - gantt chart.....	25

[Appendix – V: List of Tables]

List of Tables

Table 1- Schedule Management Plan	26
Table 2 - budget	27

List of Abbreviations

AI	Artificial Intelligent
IOT	Internet Of Things
RFID	Radio Frequency Identification
ML	Machine Learning
HTTP	Hyper Text Transfer Protocol
NIFI	NiagaraFiles
SDK	Software Development Kit
MQTT	Message Queuing Telemetry Transport
SQL	Structured Query Language.
RTSP	Real-Time Streaming Protocol

1.0 INTRODUCTION

1.1 Background and Literature

Artificial Intelligence (AI), the Internet of Things (IoT), and cloud-based technologies are being adopted at a rapid pace in the modern workplace to improve resource efficiency, employee well-being, and productivity. However, current systems frequently prioritize discrete elements—such as task scheduling or stress monitoring—instead of incorporating them into a cohesive smart office system. Inspired by previous research, this study addresses these gaps and proposes an AI and IoT-enabled Smart Office System.

Previous studies emphasize the role of vision-based and multimodal monitoring systems in workplace environments. For instance, Jeon et al. [1] developed a vision-based motion prediction system for real-time worker safety, highlighting the potential of camera-driven analytics in employee monitoring. Similarly, Arun Kumar and Prabavathy [7] proposed a hybrid multimodal AI framework for real-time activity recognition, proving the effectiveness of AI and IoT in workplace motion analytics. These studies demonstrate the feasibility of integrating IoT cameras and AI algorithms for generating employee activity reports.

Work performance and scheduling optimization have also been key research areas. Özdenizci Köse et al. [2] presented a scheduling system to enhance and monitor employee performance, while Malik et al. [10] designed an adaptive emergency scheduling algorithm for task management in IoT systems. The objectives of the proposed Smart Task Scheduling & Monitoring System are in keeping with both studies' emphasis on how AI-driven scheduling frameworks can increase accountability and expedite processes.

Employee stress and wellness management have gained increasing attention in literature. Kooistra [3] explored the use of AI-generated adaptive auditory environments to reduce stress in shared workplaces, while Mandala et al. [4] focused on stress detection through wearable heart-rate devices. Complementing this, Revathi et al. [5] and Aditya [8] highlighted the broader applications of AI-driven stress relief and adaptation during organizational changes. Ramya et al. [9] also introduced a stress management system targeting workplace environments. Collectively, these works underscore the necessity of real-time stress monitoring systems for enhancing employee well-being.

Moreover, Sharma et al. [6] provided insights into the latest AI-based intelligent sensing trends, showcasing how IoT sensors can be leveraged for continuous monitoring in dynamic environments. These findings support the integration of sensor-based and AI-driven stress detection in the proposed system. While these studies offer valuable contributions, they remain domain-specific, focusing either on stress detection [3,4,5,8,9], task scheduling [2,10], or employee activity monitoring [1,7]. Few attempts have been made to integrate these elements into a holistic smart office solution. Therefore, this research aims to bridge the gap by designing a comprehensive system that simultaneously addresses automated reporting, stress management, and intelligent scheduling, enabling data-driven decision-making for both management and employees.

1.2 Research Gap

Stress Smart office systems powered by AI and IoT are cutting-edge technologies that have enormous potential to raise worker happiness, productivity, and overall workplace well-being. Prior to such systems being widely used in practice, a number of research gaps must be filled. A primary obstacle is the absence of integration; the majority of current solutions concentrate on either job scheduling, stress detection, or staff monitoring, but rarely integrate all of these elements into a single, cohesive system.

Another challenge is analyzing and interpreting data in real time. Although stress detection and task management have been studied independently, few systems offer ongoing, real-time monitoring that connects job delegation, operational effectiveness, and employee well-being. There are still issues of ethics and practicality. Sensitive data must be handled carefully and privacy concerns are raised by ongoing camera and sensor monitoring. Additionally, the efficacy and precision of AI-powered stress detection and task scheduling in actual work environments must be confirmed.

There are also issues of ethics and practicality. Sensitive data must be handled carefully, and privacy concerns are raised by continuous camera and sensor monitoring. Additionally, the efficacy and precision of AI-powered stress detection and task scheduling in actual work environments must be validated.

To address these challenges, this study will gather and analyze data in an actual office setting. The ultimate aim is to create a complete, integrated smart office solution that enhances employee well-being and productivity while being practical and adaptable to real-world conditions.

1.3 Research Problem

As digital technologies are quickly being incorporated into modern workspaces, businesses are turning more and more to cloud-based, AI, and IoT solutions to boost efficiency and productivity. Although there are many benefits to modern technologies, employees frequently experience high levels of stress as a result of strict deadlines, hefty workloads, and the expectation to perform consistently. Stress influences not only the health of employees but also the general productivity and decision-making of the firm.

Existing workplace monitoring systems are capable of tracking task completion and attendance, but they hardly ever offer a comprehensive picture that connects management of tasks, stress levels, and employee activities. Stress detection is not linked to job scheduling, and most systems operate in isolation. Because of this disarray, it is challenging for businesses to simultaneously manage workplace resources, assign tasks efficiently, and identify stressed workers in real time.

Continuous monitoring with AI and IoT also presents ethical and privacy issues, and many current systems are not equipped to manage real-time data collecting and processing efficiently. A cohesive strategy is necessary for firms to make data-driven decisions that enhance operational effectiveness and employee well-being.

Therefore, the goal of this research is to design a smart office system that combines AI and IoT to improve workplace productivity and wellness. This system will be able to automate activity reporting, optimize task scheduling, and detect employee stress in real time.

2.0 Objectives

2.1 Main Objective

To design and develop an AI and IoT-enabled Smart Office System that enhances workplace productivity, monitors employee stress in real-time, automates activity reporting, and optimizes task scheduling in a holistic and integrated manner.

2.2 Specific Objectives

The following are the sub-objectives of conducting this research.

- Real-Time Employee Activity Monitoring.
- AI-Based Stress Detection.
- Automated Reporting System
- Smart Task Scheduling & Monitoring
- Data Integration and Decision Support

3.0 METHODOLOGY

3.1 Requirement Gathering

The most important and initial stage in creating a smart office system is gathering requirements. Employees, managers, and IT administrators are among the stakeholders whose requirements and expectations must be determined during this phase. The system's functional and non-functional needs must be defined precisely so that it can handle workplace challenges effectively in real-world scenarios.

3.1.1 Past Research Analysis

It is crucial to examine previous studies to understand current solutions, identify their shortcomings, and guide the development of a new, integrated smart office system. Task management, stress detection, and personnel monitoring are the three primary project-related areas analyzed in this study.

Employee Monitoring: To monitor employee activities, prior research has looked into IoT-enabled devices and vision-based systems. These technologies efficiently keep an eye on interest levels, attendance, and movements. Nevertheless, the majority of implementations concentrate on safety or tracking specific activities without incorporating the data into platforms for automated reporting or decision-making.

Stress Detection: Studies have indicated that wearable technology, heart rate monitoring, and facial recognition can all be used by AI and IoT to identify stress. Even while these systems are good at detecting stress signals, they frequently don't integrate with workload and task management in real time, which restricts their applicability in terms of enhancing worker productivity and wellbeing in general.

Task Scheduling : AI-powered task management platforms have been developed to allocate jobs intelligently, streamline workflows, and monitor performance. However, these solutions typically operate independently, without integrating employee activity data or stress monitoring for a unified, intelligent system. to enable integrated, intelligent decision-making.

From this analysis, it is evident that while significant progress has been made in task scheduling, stress detection, and employee monitoring, no comprehensive solution currently integrates these features. The proposed smart office system aims to fill this gap by combining AI-based employee activity reporting, real-time stress monitoring, and intelligent task scheduling into a single, unified platform.

3.1.2 Identifying Existing Systems

- Analyzing current solutions and their features is crucial to designing a successful smart office system. One of the following topics is usually the Analyzing current solutions and their features is crucial to designing a successful smart office system. One of the following topics is usually the focus of current workplace monitoring and management systems:

Employee Activity Monitoring: Attendance, mobility, and engagement are tracked via camera-based systems and IoT-enabled devices. Although they work well for basic monitoring, they typically don't integrate with task scheduling or stress detection.

Stress Detection Systems: AI-powered solutions that use wearable technology, heart rate sensors, or facial recognition can instantly detect employee stress. Nonetheless, these systems often function autonomously and do not contribute to task management or reporting systems.

job Scheduling Systems: AI-powered workflow management and job assignment systems maximize efficiency and track task completion.

3.2 Feasibility Study

3.2.1 Technical Feasibility

3.2.1.1 Knowledge on Technologies

IoT, AI/ML, data processing, cloud computing, and visualization tools are just a few of the technologies that are necessary for the AI-Based Employee Current Report Generator System with Camera, AI-Based Smart Live Employee Stress Management System, and AI-Based Smart Task Scheduling & Monitoring System. The main technical knowledge areas for each system are described below:

1. AI-Based Employee Current Report Generator System using Camera

- **IoT and Hardware Knowledge:**
 - Proficiency in configuring and managing IoT-enabled cameras (e.g., Raspberry Pi cameras, Hikvision IP cameras) and sensors (e.g., motion detectors, RFID tags) for real-time data capture.
 - Understanding of network protocols (e.g., MQTT, HTTP) to ensure seamless data transmission from devices.
- **AI and Machine Learning:**
 - Expertise in computer vision frameworks like OpenCV for video processing and object detection.
 - Ability to develop and train deep learning models using TensorFlow or PyTorch, including YOLO for real-time employee activity recognition.
- **Data Processing and Analytics:**
 - Knowledge of real-time data streaming tools like Apache Kafka to handle large volumes of video and sensor data.
 - Skills in Python programming for integrating AI models with hardware inputs.
- **Visualization and Reporting:**
 - Familiarity with business intelligence tools such as Power BI or Tableau for creating actionable reports from AI-processed metrics.
- **Cloud Computing:**
 - Understanding of AWS IoT or Google Cloud IoT for device management and Azure Machine Learning for deploying scalable AI models.

2. AI-Based Smart Live Employee Stress Management System

- IoT and Hardware Knowledge:
 - Expertise in integrating wearable sensors (e.g., Fitbit, Empatica E4) and IoT cameras (e.g., Logitech C920) to collect physiological and behavioral data.
 - Knowledge of sensor data formats and calibration techniques for accurate stress detection.
- AI and Machine Learning:
 - Proficiency in TensorFlow or Keras for building models to analyze stress indicators from sensor data and facial expressions via OpenCV.
 - Understanding of Scikit-learn for classifying stress levels based on real-time inputs.
- Data Processing:
 - Skills in Apache Spark for processing large-scale sensor data streams and Node-RED for integrating diverse data sources.
- Intervention and Communication:
 - Familiarity with collaboration tools like Slack or Microsoft Teams for real-time alerts and mobile app development (e.g., Flutter) for employee feedback.
- Cloud Computing:
 - Knowledge of Google Cloud AI for model deployment and IBM Watson Health for health data analytics and wellness insights.

3. Smart Task Scheduling & Monitoring System

- IoT and Hardware Knowledge:
 - Proficiency in deploying IoT sensors and wearables to track employee availability and workload.
 - Understanding of IoT data collection and transmission protocols (e.g., MQTT, CoAP).
- AI and Machine Learning:
 - Expertise in OptaPlanner for constraint-based task optimization and TensorFlow for predicting employee performance trends.
 - Knowledge of reinforcement learning for adaptive task assignment.
- Data Processing and Storage:
 - Skills in Apache NiFi for data routing and PostgreSQL for managing historical task data.

- Task Management and Monitoring:
 - Familiarity with project management tools like Asana, Trello, or Jira, integrated with AI-driven scheduling.
- Cloud Computing:
 - Understanding of AWS Lambda for serverless scheduling execution, Microsoft Azure IoT Hub for real-time data processing, and Google Kubernetes Engine for system scalability.

Internet of Things (IoT): IoT makes it possible to gather data in real time from a variety of wearable technology, sensors, and cameras placed across the office. These gadgets track the physiological markers, engagement, and activity of employees. In order to make the system sensitive and adaptable to real-world circumstances, IoT makes sure that data flows continuously to the AI modules for processing.

Cloud technology: Cloud infrastructure offers remote accessibility, data synchronization, and centralized storage. It makes it possible for gathered data to be safely kept, processed quickly, and made available to management or authorized staff from any place. Moreover, real-time updates, automated reporting, and module integration are made easier by cloud connectivity.

3.2.1.2 Knowledge on Tools

The successful implementation of the suggested systems necessitates expertise in a variety of specialized tools for cloud management, data processing, AI/ML development, IoT integration, and visualization. The essential tools connected to each system are described as follows:

1. AI-Based Employee Current Report Generator System using Camera

- IoT Tools:
 - Raspberry Pi Camera Module or Hikvision IP Cameras: For capturing real-time video and image data from the workplace.
 - Motion Sensors/RFID Readers: For tracking attendance and movement (e.g., Adafruit RFID readers).
- AI/ML Development Tools:
 - OpenCV: For video processing, object detection, and tracking employee

activities.

- TensorFlow: For training custom deep learning models to analyze attendance and task engagement.
- YOLO: A pre-trained model integrated with OpenCV for real-time object recognition.
- Data Processing Tools:
 - Apache Kafka: For streaming and managing high-volume camera data in real-time.
 - Python (with Libraries like NumPy, Pandas): For scripting and data manipulation.
- Visualization and Reporting Tools:
 - Power BI: For creating interactive dashboards and reports from AI-generated metrics.
 - Tableau: For advanced data visualization and sharing insights with management.
- Cloud Tools:
 - AWS IoT Core: For connecting and managing IoT devices securely.
 - Azure Machine Learning: For deploying and scaling AI models on the cloud.
 - Google Cloud Storage: For storing processed video and report data.

2. AI-Based Smart Live Employee Stress Management System

- IoT Tools:
 - Fitbit SDK or Empatica E4: For collecting physiological data (heart rate, skin temperature) from wearables.
 - Logitech C920 with IoT Integration: For capturing facial expressions via camera feeds.
- AI/ML Development Tools:
 - TensorFlow/Keras: For building and training models to detect stress from sensor and video data.
 - OpenCV: For facial recognition and emotion analysis.
 - Scikit-learn: For implementing machine learning classifiers to identify stress patterns.
- Data Processing Tools:
 - Apache Spark: For processing and analyzing large-scale sensor data streams.

- Node-RED: For creating data flow integrations between sensors, AI models, and outputs.
- Intervention and Communication Tools:
 - Slack API: For sending real-time stress alerts to managers or HR.
 - Flutter: For developing a mobile app to provide stress-relief prompts to employees.
- Cloud Tools:
 - Google Cloud AI Platform: For deploying pre-trained models and real-time inference.
 - IBM Watson Health: For health data analysis and generating wellness recommendations.
 - AWS S3: For storing historical stress and intervention data.

3. Smart Task Scheduling & Monitoring System

- IoT Tools:
 - RFID Readers or Bluetooth Beacons: For tracking employee location and availability (e.g., Kontakt.io beacons).
 - Garmin Connect SDK: For integrating workload and activity data from smartwatches.
- AI/ML Development Tools:
 - OptaPlanner: For optimizing task scheduling based on constraints like workload and availability.
 - TensorFlow: For predicting employee performance and availability trends.
 - OpenAI Gym: For implementing reinforcement learning to adapt task assignments dynamically.
- Data Processing Tools:
 - Apache NiFi: For routing and transforming IoT data into the scheduling system.
 - PostgreSQL: For storing and querying historical task and performance data.
- Task Management and Monitoring Tools:
 - Asana API: For task assignment and tracking, integrated with AI outputs.
 - Trello Power-Ups: For visualizing task progress in real-time.
 - Jira Software: For project management and deadline monitoring.
- Cloud Tools:

- AWS Lambda: For running serverless scheduling algorithms on demand.
- Microsoft Azure IoT Hub: For collecting and processing IoT data in real-time.
- Google Kubernetes Engine: For orchestrating and scaling the monitoring system.

3.2.1.3 Data collection Knowledge

A thorough grasp of data sources, acquisition strategies, preprocessing techniques, and management practices are necessary for efficient data gathering, which is essential to the operation of the suggested systems. The understanding of data collecting required for each system is outlined below:

1. AI-Based Employee Current Report Generator System using Camera

- Data Sources:
 - Video Feeds: Real-time video from IoT-enabled cameras (e.g., Raspberry Pi, Hikvision) capturing employee activities.
 - Sensor Data: Motion sensors or RFID tags for attendance and movement tracking.
- Collection Methods:
 - Continuous Streaming: Use of RTSP or HTTP for live video capture.
 - Event-Triggered Sampling: Collect sensor data (e.g., RFID scans) when employees enter/exit designated areas.
- Data Formats:
 - Video: MP4 or H.264 formats for efficient storage and processing.
 - Sensor Data: JSON or CSV files for structured attendance and movement logs.
- Preprocessing Knowledge:
 - Frame Extraction: Use OpenCV to extract key frames from video for AI analysis.
 - Noise Reduction: Apply filters to remove irrelevant movements or lighting artifacts.
- Challenges and Solutions:
 - Privacy Concerns: Implement anonymization techniques (e.g., blurring faces) and comply with GDPR.
 - Bandwidth: Optimize compression (e.g., H.265) to manage high-volume video data.
 - Storage: Use cloud storage (e.g., AWS S3) with tiered retention policies.

2. AI-Based Smart Live Employee Stress Management System

- Data Sources:
 - Sensor Data: Physiological data from wearables (e.g., Fitbit, Empatica E4) including heart rate, skin temperature, and activity levels.
 - Video Data: Facial expressions and body language from IoT cameras (e.g., Logitech C920).
- Collection Methods:
 - Real-Time Streaming: Use Bluetooth Low Energy (BLE) or MQTT for sensor data and RTSP for video feeds.
 - Periodic Sampling: Collect sensor data at regular intervals (e.g., every 5 seconds) to balance accuracy and battery life.

- Data Formats:
 - Sensor Data: Time-series data in CSV or HDF5 formats for efficient analysis.
 - Video Data: MP4 with metadata tags for timestamped stress events.
- Preprocessing Knowledge:
 - Data Synchronization: Align sensor and video data using timestamps to correlate physiological and behavioral indicators.
 - Normalization: Scale sensor readings (e.g., heart rate to 0-1 range) for consistent AI input.
- Challenges and Solutions:
 - Data Accuracy: Calibrate sensors regularly and validate against manual observations.
 - Employee Consent: Obtain informed consent and provide opt-out options for data collection.
 - Latency: Use edge computing to process data locally before cloud transmission.

3. Smart Task Scheduling & Monitoring System

- Data Sources:
 - IoT Data: Location and availability from RFID readers, Bluetooth beacons, or wearables (e.g., Garmin).
 - Performance Data: Historical task completion times and productivity metrics from project management tools (e.g., Jira).
- Collection Methods:
 - Passive Tracking: Continuously collect location data via RFID or BLE signals.
 - Active Reporting: Employees log task start/end times via mobile apps or APIs.
- Data Formats:
 - IoT Data: JSON or XML for real-time location and availability updates.
 - Performance Data: CSV or database tables (e.g., PostgreSQL) for historical records.
- Preprocessing Knowledge:
 - Data Aggregation: Combine location and performance data to assess workload distribution.
 - Outlier Detection: Identify and correct erroneous data (e.g., missed RFID scans) using statistical methods.
- Challenges and Solutions:
 - Data Gaps: Implement redundancy with multiple sensors to ensure continuous coverage.
 - Scalability: Use distributed data collection with Apache NiFi to handle large teams.
 - Security: Encrypt data in transit (e.g., TLS) and at rest to protect employee information.

3.2.2 Schedule Feasibility

Determining if the suggested system can be created and put into use within the allotted time is known as schedule feasibility. Requirement analysis, system design, hardware configuration, software development, integration, testing, and final deployment are some of the processes. A Gantt chart was created to show the task relationships and timeline.

3.3 System Analysis

3.3.1 Software Solution Approach

Three main modules—Current Report Generator, Stress Management, and Task Scheduling & Monitoring—are integrated into the AI-Based Employee Monitoring and Management System software solution through a modular, cloud-based architecture that guarantees scalability, real-time processing, and smooth data integration. Using AI frameworks (like TensorFlow) and computer vision libraries (like OpenCV), the Current Report Generator processes video feeds from Internet of Things cameras and uses activity detection and facial recognition algorithms to provide real-time attendance and engagement reports. In order to provide real-time stress detection and alerts over a RESTful API, the Stress Management System analyzes biometric data from sensors (such as heart rate and posture) and camera-based cues (such as facial expressions) using machine learning models (such as scikit-learn and PyTorch). With the help of AI-driven optimization algorithms (like reinforcement learning) and IoT data streams, the Task Scheduling & Monitoring System assigns assignments according to employee workload and performance indicators. It is controlled by a centralized database (like MongoDB or PostgreSQL). Agile development is used to create the system, and microservices are set up on cloud platforms (like AWS or Azure) to provide scalability and fault tolerance. Management accessibility is ensured via a uniform dashboard that is constructed with React or Angular and offers real-time visualization and control. This strategy guarantees strong integration, scalability, and flexibility; yet, issues like data privacy and computational expenses necessitate cautious handling through encryption and well-designed algorithms.

AI-Based Employee Current Report Generator System using Camera flowchart

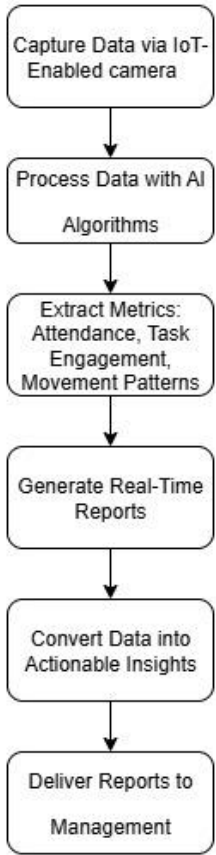


Figure 3-current report generator system flowchart

AI-Based Smart Live Employee Stress Management System

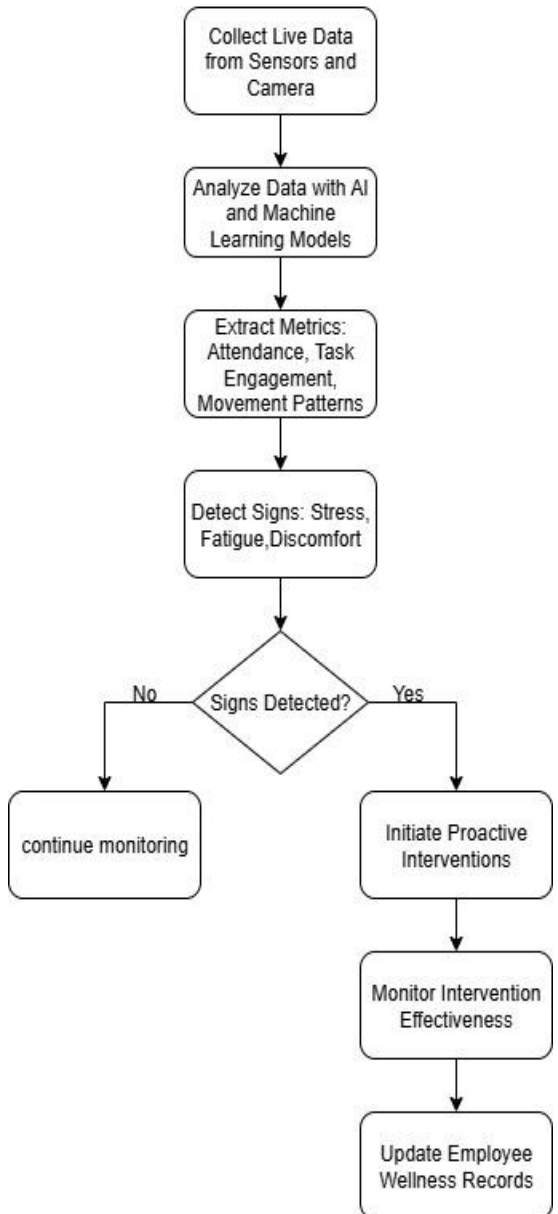


Figure 2 2 - Live employee stress management system

Smart Task Scheduling & Monitoring System

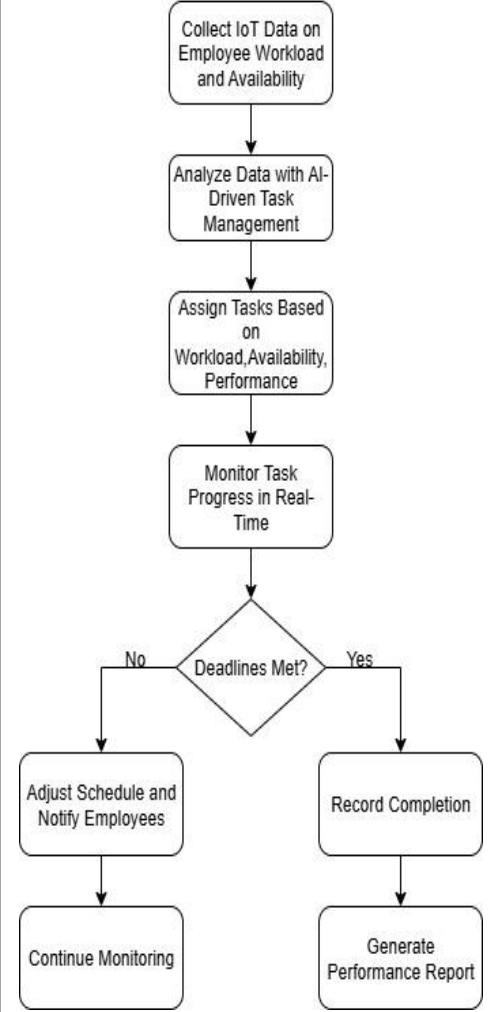


Figure 1 3 - task scheduling & monitoring system flowchart

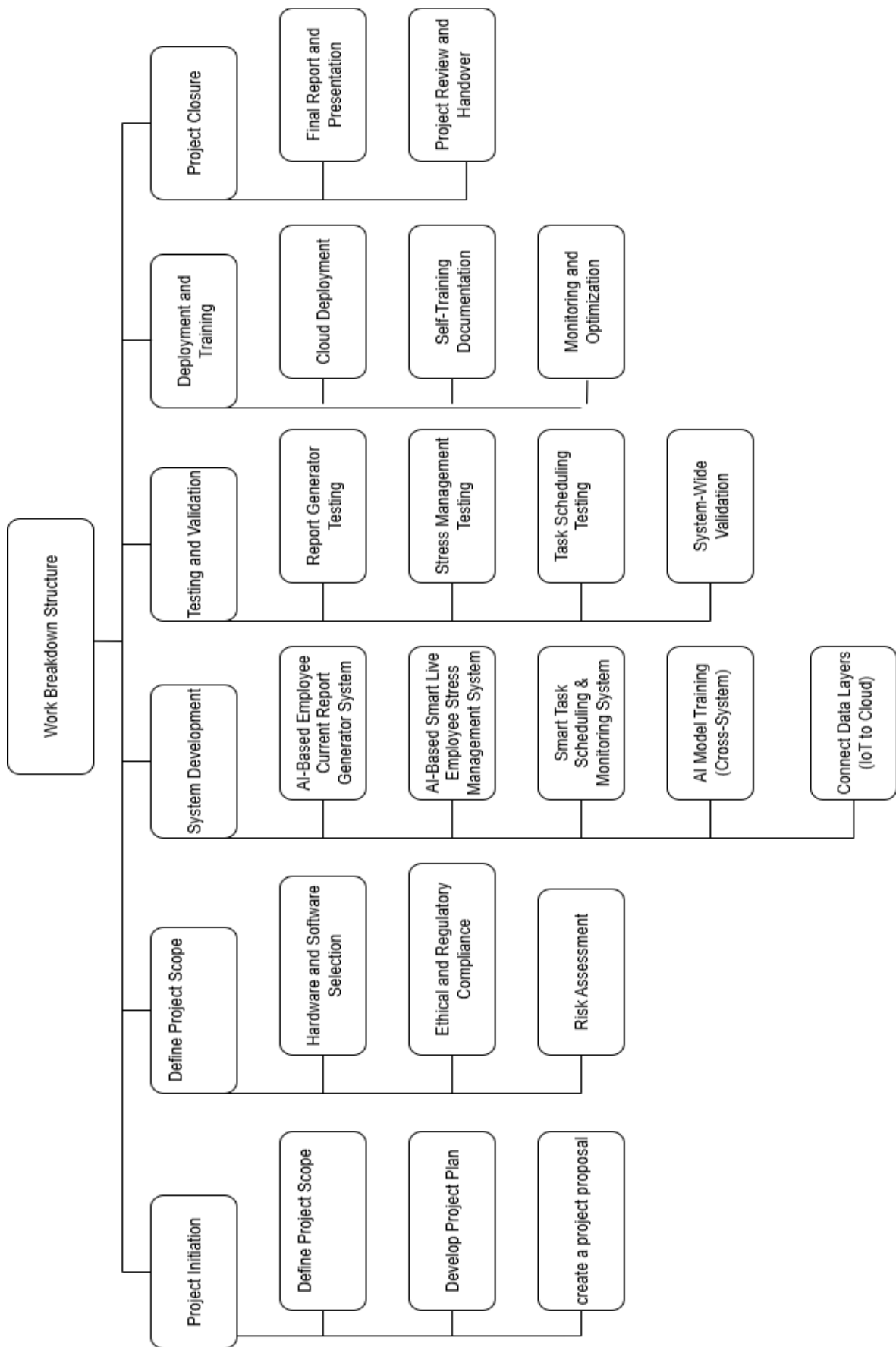


Figure 4 - work breakdown structure

3.4 Project Requirements

3.4.1 Functional Requirements

The AI-Based Employee Monitoring and Management System requires specific functionalities to achieve its objectives:

- **Current Report Generator:** Must capture real-time data via IoT cameras, perform facial recognition and activity detection using AI algorithms (e.g., OpenCV, YOLO), and generate automated reports on attendance, task engagement, and movement patterns, exportable in formats like PDF and CSV.
- **Stress Management System:** Must analyze live sensor data (e.g., heart rate, posture) and camera-based cues (e.g., facial expressions) using machine learning models (e.g., PyTorch), detect stress or fatigue, and trigger real-time alerts or interventions for management.
- **Task Scheduling & Monitoring System:** Must integrate IoT data to assign tasks based on employee workload, availability, and performance metrics, using AI-driven optimization (e.g., reinforcement learning), and provide real-time progress tracking via a dashboard.
- **Unified Dashboard:** Must integrate all modules, offering real-time visualization of reports, stress alerts, and task statuses, with user-friendly interfaces for managers to access insights and control settings.

3.4.2 Non-Functional Requirements

The system must meet performance, security, and usability standards to ensure reliability and adoption:

- **Performance:** Process real-time data with latency under 2 seconds for report generation and stress detection, and support scalability for up to 500 employees across multiple sites.
- **Reliability:** Achieve 99.9% uptime for cloud-based services (e.g., AWS EC2) and ensure fault tolerance through redundant systems.
- **Security:** Implement AES-256 encryption for data transmission and storage, comply with data protection regulations (e.g., GDPR), and use OAuth 2.0 for secure user authentication.
- **Usability:** Provide an intuitive React-based dashboard with minimal training required (under 2 hours) for managers to navigate and interpret data.
- **Interoperability:** Integrate seamlessly with existing HR systems (e.g., SAP, Workday) via RESTful APIs and support MQTT for IoT data streaming.
- **Maintainability:** Use modular microservices architecture (e.g., Docker, FastAPI) to facilitate updates and maintenance, with automated testing via Jenkins.

3.5 Testing

The AI-Powered Employee Monitoring and Management System satisfies both functional and non-functional needs. In order to validate individual components, unit testing aims for >95% facial recognition accuracy (OpenCV/YOLO), >90% sensitivity for stress detection (PyTorch), and <5% scheduling conflicts (Stable-Baselines3). Integration testing ensures that data flows smoothly over MQTT and RESTful APIs. System testing confirms GDPR compliance, AWS 99.9% uptime, and scalability for 500 employees. With less than two hours of training, user acceptance testing guarantees that the React dashboard is user-friendly. Performance testing aims for a delay of less than two seconds, while security testing verifies AES-256 encryption and OAuth 2.0. Defects are addressed using manual reviews and automated tools (Jenkins, Selenium).

3.6 Timeline

The proposed timeline for the project is as follows.

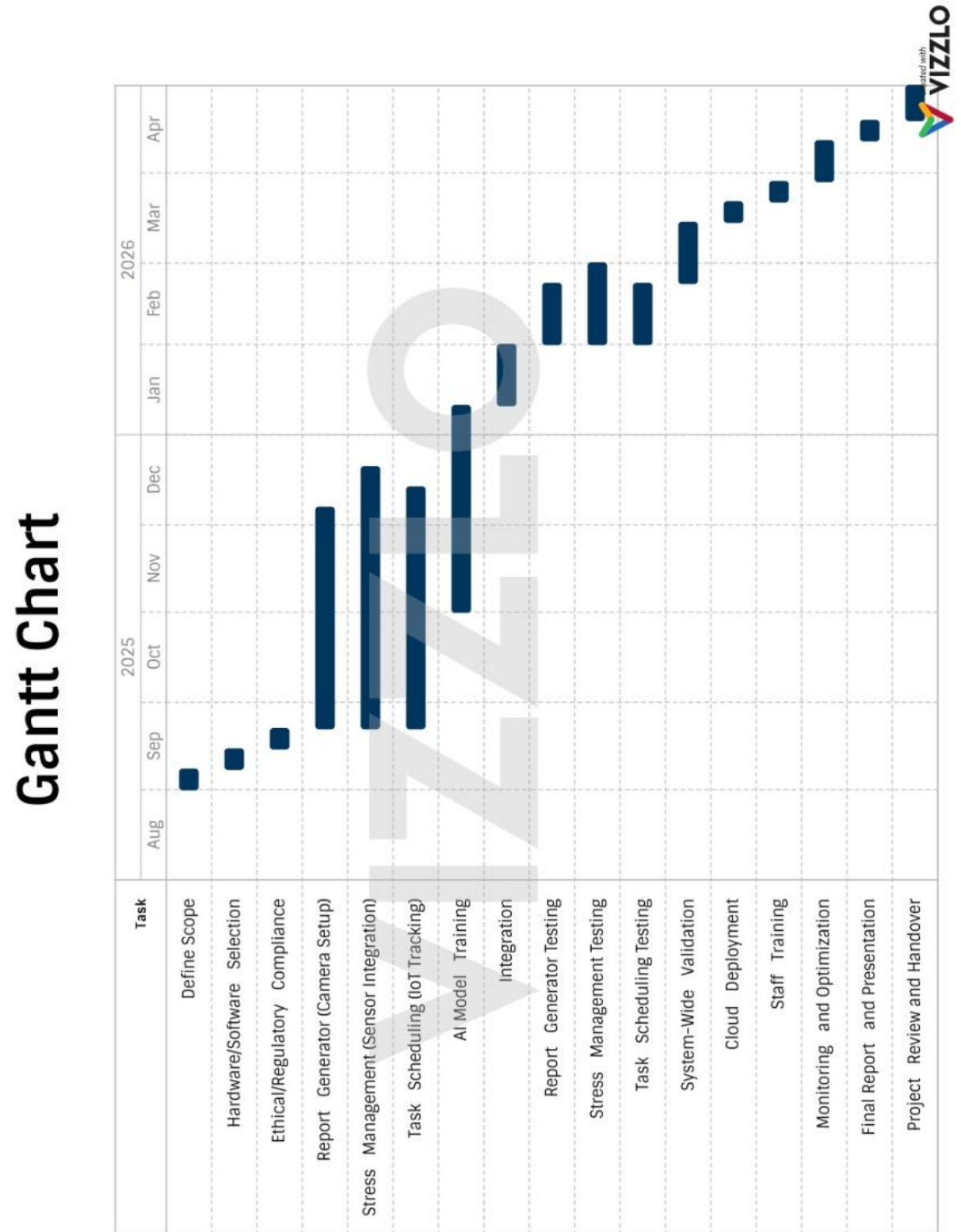


Figure 5 - gantt chart

Table 1-Schedule Management Plan

Task	Start Date	End Date	Duration (Weeks)
Define Scope	01-Sep-2025	07-Sep-2025	1
Hardware/Software Selection	08-Sep-2025	14-Sep-2025	1
Ethical/Regulatory Compliance	15-Sep-2025	21-Sep-2025	1
Report Generator (Camera Setup)	22-Sep-2025	06-Dec-2025	11
Stress Management (Sensor Integration)	22-Sep-2025	20-Dec-2025	13
Task Scheduling (IoT Tracking)	22-Sep-2025	13-Dec-2025	12
AI Model Training	01-Nov-2025	10-Jan-2026	10
Integration	11-Jan-2026	31-Jan-2026	3
Report Generator Testing	01-Feb-2026	21-Feb-2026	3
Stress Management Testing	01-Feb-2026	28-Feb-2026	4
Task Scheduling Testing	01-Feb-2026	21-Feb-2026	3
System-Wide Validation	22-Feb-2026	14-Mar-2026	3
Cloud Deployment	15-Mar-2026	21-Mar-2026	1
Staff Training	22-Mar-2026	28-Mar-2026	1
Monitoring and Optimization	29-Mar-2026	11-Apr-2026	2
Final Report and Presentation	12-Apr-2026	18-Apr-2026	1
Project Review and Handover	19-Apr-2026	30-Apr-2026	2

4.0 Budget

The projected expenses needed to develop and deploy the suggested AI and IoT-enabled Smart Office System are listed in the budget. In addition to software tools and cloud services required for data processing, monitoring, and reporting, it has hardware elements including cameras, sensors, microcontrollers, and auxiliary accessories. To guarantee system scalability, cost-effectiveness, and dependability, every component is carefully chosen.

Table 2 - budget

Item	Description	Cost (LKR)
ESP32 Microcontroller	Central control unit for IoT integration	3,500
Camera Module	IoT-enabled camera for real-time activity monitoring	6,000
Heart Rate Sensor	For AI-based stress detection	2,500
Temperature Sensor	To capture environmental data (office comfort monitoring)	1,800
Motion Sensor (PIR)	For employee activity and presence detection	1,500
Relay Module	For switching control of connected devices	1,200
Power Supply Unit	Provides stable power for sensors and microcontroller	2,000
Cloud Subscription	Firebase/AWS IoT monthly subscription	5,000
Software Tools	AI model development & data processing tools (licensing/utility cost)	4,000
Miscellaneous	Breadboards, jumper wires, resistors, casing, maintenance, and backups	3,000
	Total estimate cost	30,500

5.0 SUMMARY

The AI-Based Employee Monitoring and Management System combines three essential modules to improve worker well-being and workplace productivity: Task Scheduling & Monitoring, Stress Management, and Current Report Generator. The system makes use of AI algorithms and IoT-enabled cameras to automate real-time activity and attendance reporting, identify signs of weariness and stress for preventative measures, and assign tasks optimally based on workload and performance indicators. Using PyTorch and scikit-learn for stress analysis, OpenCV and YOLO for facial recognition, and Stable-Baselines3 for scheduling, along with a React-based dashboard on AWS for unified management, a cloud-based, modular design guarantees scalability. Non-functional criteria guarantee low latency, high uptime, GDPR-compliant security with AES-256 and OAuth 2.0, integration with HR systems, and automatic report production, stress alerts, intelligent job scheduling, and an intuitive dashboard. Using automated technologies like Jenkins and Selenium, thorough testing verifies usability, scalability, accuracy (>95% for facial recognition, >90% for stress detection), and performance. Although early cost management and data privacy compliance are problems, the system is commercially viable and offers significant productivity improvements and savings through automation. If scalability and regulatory requirements are adequately addressed during deployment, this creative approach offers to improve workplace productivity and streamline operations.

6.0 References

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